

Juan Pablo Valenzuela Castro

Software Engineer — Backend, Cloud & AI on GCP · Builds and operates production SaaS end-to-end (VALC Tech)

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PROFESSIONAL SUMMARY

Software engineer with 3+ years shipping cloud-native systems on Google Cloud Platform and Firebase. Through VALC Tech, my independent practice, I design, build and operate production SaaS end-to-end — AI agents with GPT function calling, Stripe recurring billing, Mexican CFDI 4.0 e-invoicing and offline-first PWAs — serving real businesses in Mexico, in parallel with my full-time role building data platforms at Monteli LLC. My graduate research applied machine learning to geospatial and multispectral imagery. Open to remote backend, cloud and AI roles.

TECHNICAL SKILLS

AI & LLMs: OpenAI API · GPT function calling / tool use · agent guardrails · RAG design · prompt engineering · AI workflow orchestration

Backend & APIs: Node.js · Python · FastAPI · REST APIs · microservices · async/event-driven architecture · Docker · CI/CD

Cloud (GCP & Firebase): Cloud Run · Pub/Sub · BigQuery · Cloud Functions · Firestore · Cloud Storage · IAM · Firebase Auth/Hosting

Product & Payments: Stripe (subscriptions, webhooks, metered billing) · CFDI 4.0 e-invoicing · WhatsApp Cloud API · multi-tenant SaaS

Frontend: TypeScript · Angular (signals, offline-first PWA) · Vue 3 · Ionic / Capacitor · SCSS

ML & Data: scikit-learn · MLP · SVM · computer vision · multispectral imaging · ETL pipelines · Redis · Linux · Git

EXPERIENCE

Founder & Software Engineer · VALC Tech (independent practice)

Jan 2026 – Present

- Independent software practice run alongside my full-time role: custom client systems and self-funded SaaS products, in production on GCP/Firebase.
- AgendaBot Pro: LLM agent in production booking real appointments over WhatsApp — GPT function calling where every tool executes server-side and the model never writes to the database.
- JBP (client, under contract): produce-sourcing cost control in production — per-trip costing, estimated vs. real margin, supplier advances and bank reconciliation across two legal entities.
- End-to-end ownership: architecture, backend, Stripe billing, CFDI 4.0 e-invoicing and cloud operations.

Software Engineer (Full Stack) · Monteli LLC

Jan 2023 – Present

- Designed and deployed scalable data ingestion and transformation pipelines on GCP (Cloud Run, Pub/Sub, Python, Node.js) serving multiple SaaS products and enterprise clients.
- Built event-driven microservices supporting LLM-powered workflows and large-scale automated data processing.
- Built and maintained production REST APIs for AI/ML inference services, enabling real-time processing and integration with external systems.
- Integrated the OpenAI API into production workflows: structured outputs, tool use, and orchestration of multi-step automation pipelines.
- Owned the full backend lifecycle: API design, Dockerization, CI/CD pipelines, monitoring and production support.

FEATURED PROJECTS

AgendaBot Pro — WhatsApp appointment booking SaaS (live) | GPT function calling · WhatsApp Cloud API · Stripe · CFDI · Firebase

- Multi-tenant SaaS for Mexico/LATAM where patients book, reschedule and cancel entirely from WhatsApp; AI agent with deterministic guardrails, built and operated solo.

JBP — cost control for produce sourcing (client work, in production) | Angular · Firebase · offline-first PWA · Cloud Functions

- Costs every truckload against its purchase order and explains margin leaks (shrinkage vs. quality penalty); money stored in integer cents, offline field capture, idempotent money operations.

Balltics — the platform running a real baseball league (live) | Angular · Firebase · IndexedDB · Stripe

- Multi-tenant league platform operating a real league's full season: live pitch-by-pitch scoring, public standings and box scores, offline-first capture, recurring billing.

Numa — POS with CFDI 4.0 invoicing built in (in development) | Angular · Firebase · Stripe

- Multi-tenant, offline-first POS for Mexican small businesses; CFDI 4.0 e-invoicing inside the product instead of sold as an add-on.

Sociko — gym CRM & revenue management (pre-launch, live) | Angular · Firebase · Stripe

- Collections list flags every lapsed member with their phone number next to them — recovering a single one pays for the subscription.

Nauti Test Pro — nautical exam prep (live) | Vue 3 · Ionic · Capacitor · Firebase · Stripe

- iOS/Android + PWA with a token economy: Stripe purchases unlock AES-encrypted question banks served server-side.

AgriLumens — precision agriculture ML platform (M.S. thesis) | FastAPI · Python · OpenDroneMap · Computer Vision

- ML inference over multispectral drone imagery estimating soil inorganic nitrogen and organic matter (MLP/SVM, validated against Walkley & Black lab method).

EDUCATION

M.S. Natural Resources & Environment · IPN — National Polytechnic Institute

Jan 2024 – Jan 2026

Thesis: estimation of inorganic nitrogen and organic matter in agricultural soils using multispectral imagery and machine learning models.

B.S. Software Engineering · ITSON — Sonora Institute of Technology

Aug 2019 – May 2023